

Combustion fundamentals

Class 2



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Class objectives

- › Apply the mass conservation principle in reactive systems to determine the balanced chemical reaction that describes them.
- › Introduce key concepts such as complete combustion, theoretical air.
- › Define basic parameters of the combustion analysis, air-fuel ratio, percentage of theoretical air and equivalence ratio.



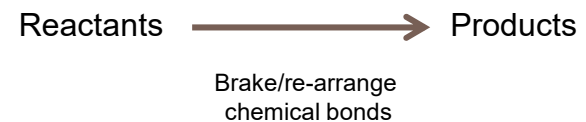
Combustion reaction



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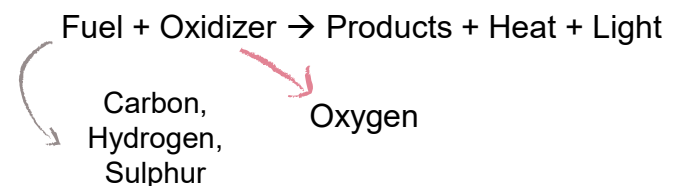
Combustion: Definition

- › What is a reactive systems?
 - Systems in which a chemical reaction takes place.



- › And combustion?
 - The transformation of the energy stored in the chemical bonds of the fuel molecules and air into heat.

- › Complete combustion
 - All combustible elements are oxidized.
- › Incomplete combustion
 - Fuel components are not burned completely
 - › C, H₂, CO, O₂ or OH in the products

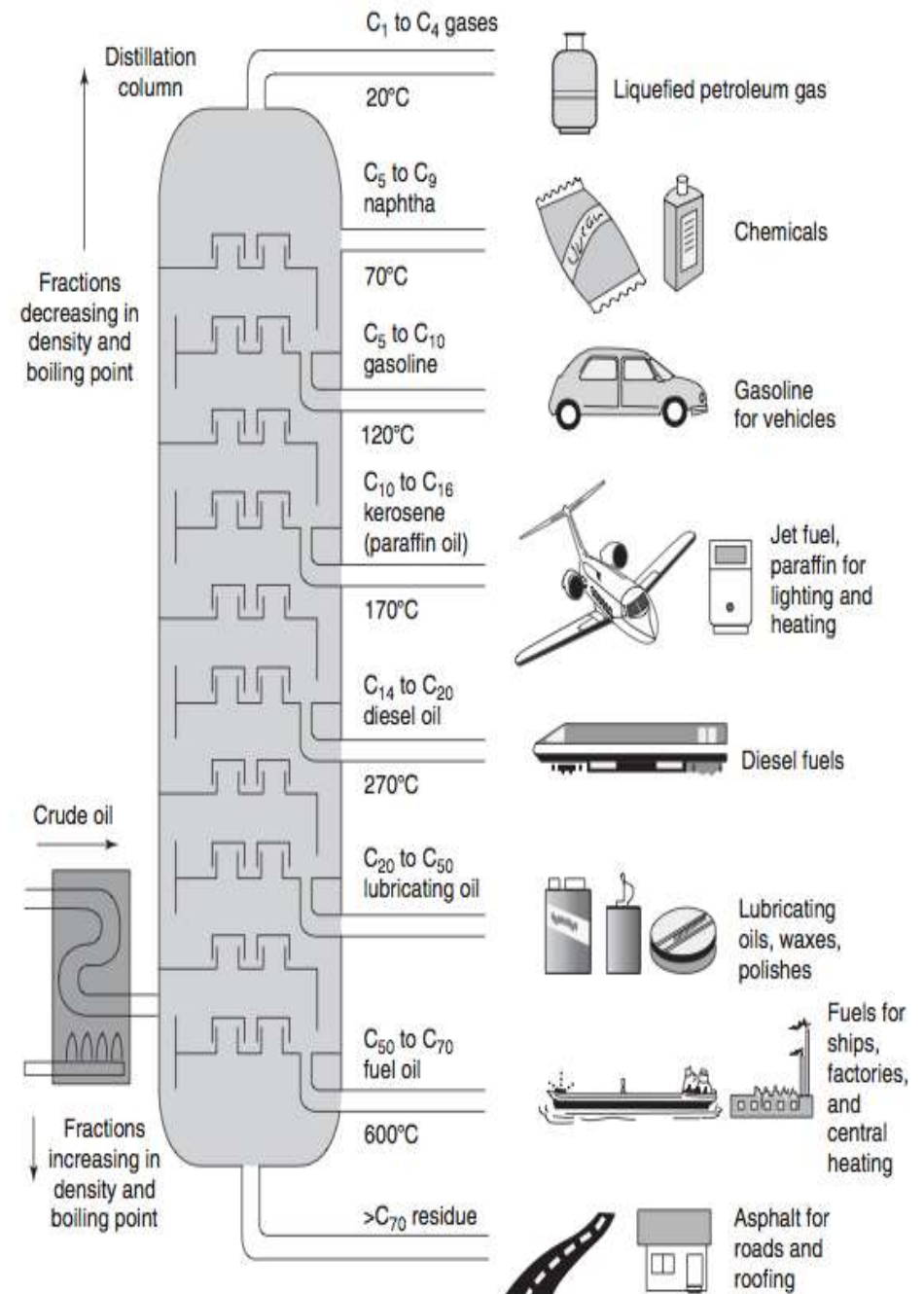


Variables

- Amount of oxygen used: Deficiency or excess inlet
- Time allowed for the combustion to take place
- Mixing: reactants must come in contact to ignite
- Elevated temperatures/pressures: molecule dissociation

FUELS

- Substance that releases heat when it reacts with and oxidizer
- Normally hydrocarbons
- Alternatives
 - Alcohols
 - Gases
 - Hydrogen
 - Naphtalines

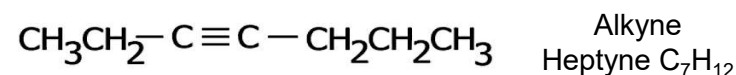
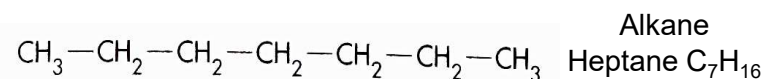


(a) Schematic diagram



Fuels: Chemistry of Hydrocarbons

- › Hydrocarbons (C_nH_m)
 - Differentiated by their bonds
- › Naming the compound:
 - Prefix - the number of carbons
 - Suffix - family to which fuel belongs



Naming convention - prefixes

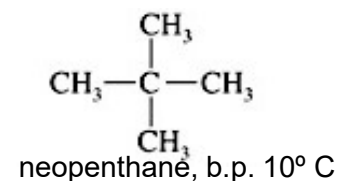
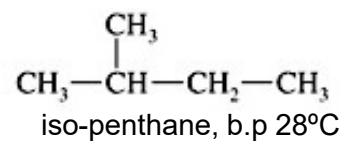
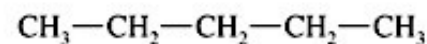
# C (n)	Prefix
1	Meth-
2	Eth-
3	Prop-
4	But-
5	Pent-

Naming conventions for hydrocarbon fuels commonly used

Family	Suffix	Formula	Bond	Example
Alkanes	-ane	C_nH_{2n+2}	Simple	Ethane
Alkenes	-ene	C_nH_{2n}	Double	Ethene
Alkynes	-yne	C_nH_{2n-2}	Triple	Ethyne

- › Hydrocarbons with a large number of carbons ($n > 3$)
 - Straight chains (denoted by n-)
 - Branched chains
 - › isomers

n-pentane, b.p 36°C



Composition of a fuel blend

- › Several fuels are actually blends.
 - Average chemical formula
 - › Component with highest concentration

General chemical composition of gasoline

60% saturated hydrocarbons	15% straight chain (n-)
	20% branched (iso-)
	15% cyclic
40% unsaturated hydrocarbons	5% alkenes
	35% aromatics
Ppm sulphur	

40 different hydrocarbons in considerable concentrations
And other in minimum concentrations

General formula for some fuels

Fuel	Average chemical formula
Gasoline	C_8H_{18}
Diesel	$C_{12}H_{26}$
Natural gas	CH_4

Oxidant: Air

- › Most common oxidant
 - O_2 only for special applications
- › Composition of dry air model
 - 79% N_2 and 21% O_2
 - › No water vapor and contaminants
 - › Consider Ar, CO_2 and the other gases as N_2

Approximate composition of dry air

Component	Mole fraction (%)
Nitrogen (N_2)	78.08
Oxygen (O_2)	20.95
Argon (Ar)	0.93
Carbon dioxide (CO_2)	0.03
Ne, He, CH_4 and others	0.01

Idealization of dry air for combustion processes

Component	Mole fraction (%)
Nitrogen (N_2)	79.05
Oxygen (O_2)	20.95

- Therefore, if:
 - 1 mole O_2 enters $\rightarrow$ 3.76 mole N_2 enters the process
 - Or 4.76 mole of total air



Oxidant Air

- › What is the role of the nitrogen?
 - Absorbes a good portion of the released chemical energy
 - N_2 is treated as inert
 - › At high temperatures N_2 forms Nox

- › What about water in the air?
 - Moist Air
 - › Water vapor present
 - › Basic model: it is inert
 - › Reality: high enough temperatures create: H_2 , O_2 , $H^\cdot$, $O^\cdot$ and $\cdot OH$



Oxidant: Air

› Theoretical air (stoichiometric)

- Minimum air needed for complete combustion of the fuel.

- › There will be no O_2 in the products
- › All fuel is completely oxidized

theoretical air $\rightarrow$ 100% theoretical air

150 % theoretical air $\rightarrow$ 50% excess

300 % theoretical air $\rightarrow$ 200% excess

80 % theoretical air $\rightarrow$ 20% deficiency

› Percent of theoretical air

- Expresses the difference between the real and the theoretical amount of air.
- Theoretical air $\rightarrow$ 100% air

Theoretical air required is 80 kmol
If the real process uses 130% of theoretical air,
How many kmol are flowing to the process?

104 kmol

› Percent excess air

- Exceeds the theoretical need

Theoretical air required is 80 kmol
If the real process uses 130% of excess air,
How many kmol are flowing to the process?

184 kmol

› Percent deficiency of air

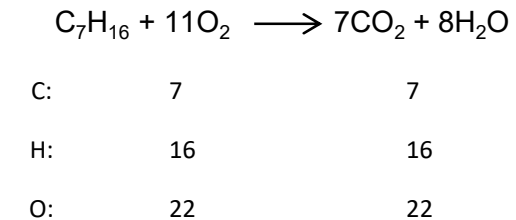
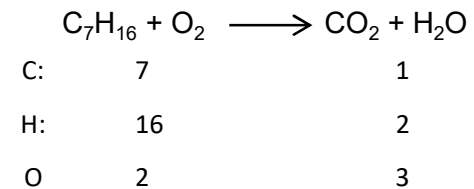
- Below the theoretical

Theoretical air required is 80 kmol
If the real process uses 79% of theoretical air,
How many kmol are flowing to the process?

63.2 kmol

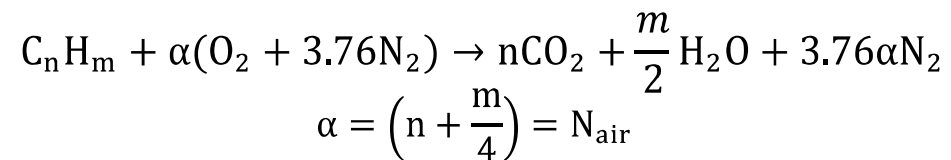
Combustion stoichiometry

- › Balanced equation
 - Mass conservation
 - › Atoms and mass are conserved
- › Stoichiometric coefficients
 - Number of moles involve



- › Balanced equation for a general hydrocarbon fuel

Fuel + Oxidizer $\rightarrow$ Products + Heat + Light



Quantifying fuel and air content

› Air-fuel ratio (AF)

- Relation between amount of fuel and air used
- Molar or mass basis

- › Expected for hydrocarbon fuels:
14-20 kg air /kg fuel

$$AF_{\text{ratio}} = \frac{m_{\text{air}}}{m_{\text{fuel}}} = \frac{4.76 * N_{\text{air}} MW_{\text{air}}}{N_{\text{fuel}} MW_{\text{fuel}}}$$

› Fuel-air ratio (FA)

- › Expected for hydrocarbon fuels:
0.05-0.07 kg fuel/kg air

$$FA_{\text{ratio}} = \frac{1}{AF_{\text{ratio}}}$$

› Equivalence ratio (ϕ)

- Relation between real and theoretical processes.

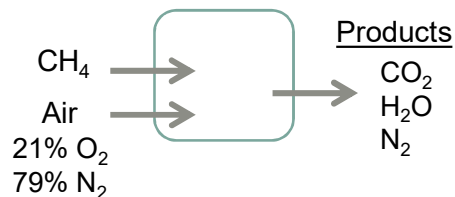
Type of mixture	
$\phi < 1$	Fuel lean mixture
$\phi = 1$	Stoichiometric mix
$\phi > 1$	Fuel rich mixture

$$\phi = \frac{AF_{\text{stoic}}}{AF_{\text{real}}} = \frac{FA_{\text{real}}}{FA_{\text{stoic}}} = \frac{m_{\text{air,stoic}}}{m_{\text{air,real}}}$$

Exercises

1. Determine the theoretical amount of air for the complete combustion of methane.

Supposition: Analysis for 1 kmol CH_4



- Balanced chemical equation for methane:



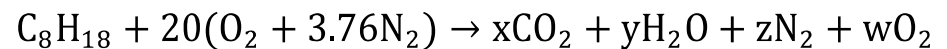
- Amount of air required: **9.52 mol air/mol methane**

$$\text{AF}_{\text{ratio}} = 17.19$$

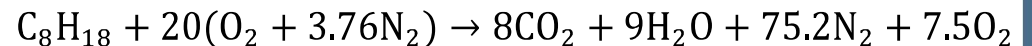
2. One kmol of octane is burned with air, which contains 20 kmol O_2 . If the products of the combustion process are only CO_2 , H_2O , O_2 and N_2 , determine a) the mole number of each of the fumes present in the products and b) the air-fuel ratio for this combustion process.

Supposition: All fumes are modelled as ideal gasses

Chemical equation:

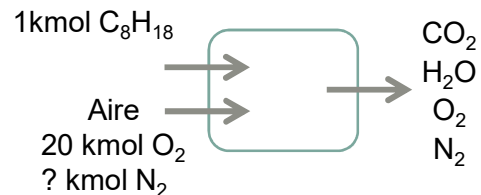


Balanced chemical equation:

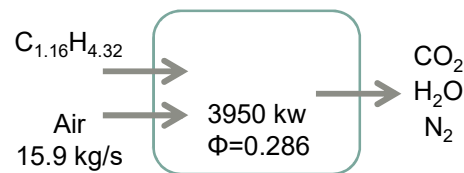


Amount of air required:

$$\text{AF}_{\text{ratio}} = 24.2 \text{ kg}_{\text{air}}/\text{kg}_{\text{fuel}}$$

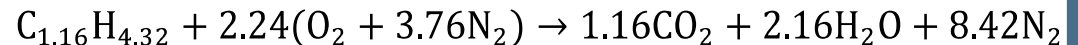


3. A small low-emission, stationary gas turbine engine operates at full load (3950 kW) at an equivalence ratio of 0.286 with an air flowrate of 15.90 kg/s. The equivalent composition of the fuel is natural gas with the chemical formula $C_{1.16}H_{4.32}$. Determine a) the fuel mass flow rate and b) the operating air-fuel ratio for the engine.



Supposition: 1) $E_k = E_p = 0$, 2) all gases are ideal gases 3) complete combustion

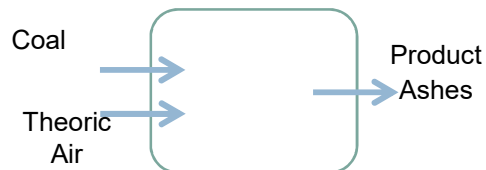
- Balanced chemical equation for ~ methane:



- Amount of air required: $AF_{stoic} = 16.82 \text{ kg}_{air}/\text{kg}_{fuel}$
- Operating AF_{real} : $AF_{real} = 59.265 \text{ kg}_{air}/\text{kg}_{fuel}$
- Fuel flowrate: $\dot{m}_{real} = 0.2683 \text{ kg}_{fuel}/\text{s}$

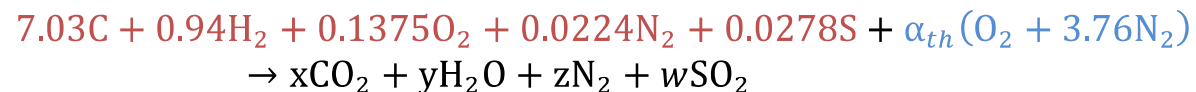
4. Pennsylvania coal, with an elemental mass analysis of 84.36% C, 1.89% H₂, 4.40% O₂, 0.63% N₂, 0.89% S and 7.83% ash, is burned with the minimum air required. If we neglect the ash, a) determine the molar fraction and the molecular weight of the combustion products. Also, calculate the AF ratio required for this combustion process.

Supposition: 1) all gases are ideal gases 2) complete combustion, products: CO₂, H₂O, SO₂, N₂
3) Total mass of coal is 100 kg

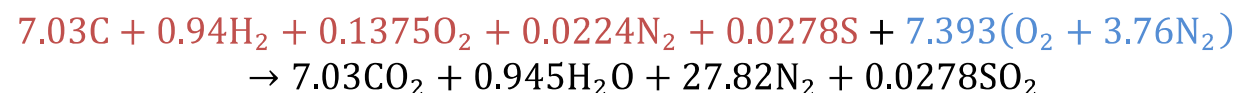


	% mass	m _i (kg)	Mw _i (kg/kmol)	N _i (kmol)
C	84.36	84.36	12	7.030
H₂	1.89	1.89	2	0.9450
O₂	4.40	4.40	32	0.1375
N₂	0.63	0.63	28	0.0225
S	0.89	0.89	32	0.0278
Tot		100		

- Chemical equation:



- Balanced chemical equation:



	N_i (kmol)	Y_i	Mw_i (kg/kmol)
CO_2	7.030	0.1962	44
H_2O	0.9450	0.0264	18
N_2	27.820	0.7766	28
SO_2	0.0278	0.000776	64
Tot	35.8228	0.9999	

- Molecular weight of the product mix:

$$MW_{\text{products}} = \frac{m_{\text{prod}}}{N_{\text{prod}}} = 30.904 \text{ kg/kmol}$$

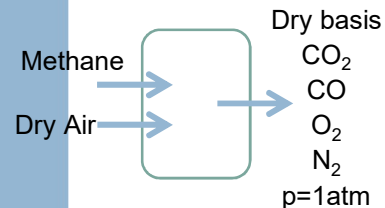
- AF ratio for the process:

$$AF_{\text{ratio}} = 10.2 \text{ kg}_{\text{air}}/\text{kg}_{\text{fuel}}$$

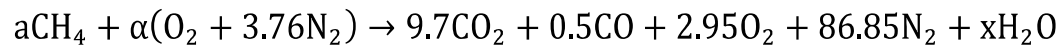
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- › Methane is burned with dry air. The molar analysis of the products on dry basis is 9.7% CO₂, 0.5%CO, 2.95%O₂, 86.85% N₂. Determine a) the air fuel ratio on both molar and mass basis, b) the percentage of theoretical air, c) the dew point temperature of the products, in F, if the mixture was cooled at 1 atm.

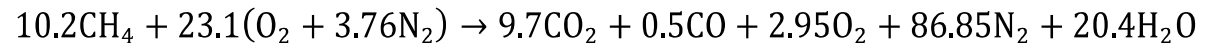
Supposition: 1) all gases are ideal gases 2) Combustion products dry basis: CO₂, CO, O₂, N₂
3) Basis for analysis 100 lbmol of dry products



- Chemical equation:

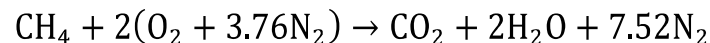


- Balanced chemical equation:



- Air-fuel ratio for the process: $\text{AF}_{\text{real}} = 10.78 \frac{\text{lbmol}_{\text{air}}}{\text{lbmol}_{\text{fuel}}} = 19.47 \frac{\text{lb}_{\text{air}}}{\text{lb}_{\text{fuel}}}$

- Chemical equation complete combustion with theoretical amount of air:



- Theoretical air-fuel ratio: $\text{AF}_{\text{stoic}} = 9.52 \frac{\text{lbmol}_{\text{air}}}{\text{lbmol}_{\text{fuel}}} = 17.19 \frac{\text{lb}_{\text{air}}}{\text{lb}_{\text{fuel}}}$

- % theoretical air-fuel:

$$\% \text{ theoretical air} = \frac{10.78 \text{ lbmol}_{\text{air}} / \text{lbmol}_{\text{fuel}}}{9.52 \text{ lbmol}_{\text{air}} / \text{lbmol}_{\text{fuel}}} = 113\%$$

- Dew point temperature:

We require the partial pressure of the water vapour (wv) present in the products.

$$y_i = \frac{p_{i,mix}}{P_{mix}} \rightarrow p_{wv,prod} = y_{wv,prod} p_{prod} = \left(\frac{N_{wv,prod}}{N_{prod}} \right) p_{prod} = 0.1694 \text{ atm}$$

We require the partial pressure in psia, since the table of saturated water properties measures pressure in those units

$$p_{wv,prod} = 0.1694 \text{ atm} = 2.4895 \frac{\text{lb}_f}{\text{in}^2}$$

Table of temperatures, saturated water.

Temp. , T(°F)	Sat. pressure; P_{sat} (psia)
110	1.2767
120	1.6951
130	2.2260
140	2.8931
150	3.7234

Interpolating from data of the table

$$T = 134^\circ\text{F}$$

If the products of combustion were cooled at constant pressure below this temperature we expect water vapour condensation in our outlet gasses.